

Week 2 Exercise Sheet

Key:

- **Questions marked with a 🍷** are for discussion in your tutorials. Please complete these questions before your tutorial.
- **Questions marked with a 🌟** are difficult questions.
- **Questions marked with a 🎵** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

🎵 **Exercise 1.** Express the following as the composition of two functions. In other words, give a f and g such that $h = g \circ f$.

- (i) $h : \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R}, h(x) = \frac{1}{x-2}$;
- (ii) $h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = \sin^4(2x)$;
- (iii) $h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = 2 \sin^3(x) + 5 \sin(x)$;
- (iv) $h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = \frac{1}{2 + \cos(x)}$.

Explain whether it is possible to find alternative answers than those you obtained.

🍷 **Exercise 2.** Express the following function rules as piecewise forms, without any absolute value signs.

- (i) $f : [0, 2\pi] \rightarrow \mathbb{R}, f(x) = |\cos(x)|$;
- (ii) $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = |x^2 - 4| + 4$;
- (iii) $h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = |x - 1| + |x| + |x + 1|$.

🍷 **Exercise 3.** Give the complete function specification (domain, codomain, rule) for the compositions $f \circ g$ and $g \circ f$ when

- (i) $f : [2, \infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x-2}$ and $g : (-\infty, 3] \rightarrow \mathbb{R}, g(x) = \sqrt{3-x}$.
- (ii) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 + 3$ and $g : [3, \infty) \rightarrow \mathbb{R}, g(x) = \sqrt{x-3}$.
- (iii) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{x^2}{1+x^2}$ and $g : (0, \infty) \rightarrow \mathbb{R}, g(x) = \ln(x)$.

🍷 **Exercise 4.** Decompose each of the following function rules as a sum of odd and even functions (simplify your answers where possible).

- (i) $\frac{1}{x+1}$;
- (ii) $\left(\frac{1-x}{1+x}\right)^{1/2}$;
- (iii) $\sin(x+1)$.

♣ **Exercise 5.** (i) Find two functions f and g such that $f \circ g = g \circ f$.

(ii) Suppose f is an odd function and $0 \in D(f)$. What can you say about $f(0)$?

(iii) What can you say about a function that is both even and odd?

♣ **Exercise 6.** Find the domain of the function rule $k(x) = \frac{|x|}{x^2}$ and then sketch it.

♣ **Exercise 7.** Let $f(x) = x^3 - x$ and $g(x) = x^{\frac{1}{4}}$. State the largest subsets of real numbers for which f and g are defined as functions.

For the following functions, state the largest subset of real numbers for which they are defined as functions:

- (i) $f + g$;
- (ii) $g \circ g$;
- (iii) $g \circ f$;
- (iv) $g \circ \frac{1}{f}$.

♣ **Exercise 8.** (i) Give two examples of even functions.

(ii) Give two examples of odd functions.

(iii) State the definition of a periodic function. Give two examples of periodic functions.

(iv) State the definitions of monotonic increasing and monotonic decreasing functions. Give two examples of each type.

♣ **Exercise 9.** (i) Find all real solutions to the equation $2 \cos^2(x) + 3 \sin(x) = 3$.

(ii) Show from the definition of the hyperbolic functions that $\cosh^2(t) + \sinh^2(t) = \cosh(2t)$.

(iii) Find all the solutions to the equation $3 \sinh(x) - \cosh(x) = 1$.

(iv) Show that for any $x \in \mathbb{R}$, $\sinh^{-1}(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$.

♣ **Exercise 10.** Suppose that f and g are functions defined for all $x \in \mathbb{R}$ and f is monotonic increasing and g is monotonic decreasing.

(i) Show that $f \circ g$ and $g \circ f$ are both monotonic decreasing.

(ii) Is $f \circ f \circ g$ monotonic increasing or decreasing?